

MANUEL PAEZ

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Education

Columbia University

New York, NY

Master of Science, Computer Science, GPA: 3.8/4.0

Jan. 2025 - Expected May 2026

Bachelor of Arts, Mathematics

Sept. 2023 - Expected May 2026

Bachelor of Arts, Computer Science

Sept. 2019 - May 2023

Relevant Coursework: Advanced Probability Theory, Algorithms for Massive Data, Analysis and Probability, Machine Learning Theory, Log-Concave Probability, Unsupervised Learning, Theoretical Foundations of LLMs.

Work and Research Experience

Flatiron Institute - Simons Foundation

New York, NY

Visiting Scholar Fellow, Neural Circuits and Algorithms Group

Nov. 2024 - Aug. 2025

- Developing biologically plausible algorithms for learning low-dimensional predictive representations of nonlinear dynamical systems using rectified and asymmetric similarity matching. Code predominantly in Python.

Columbia University

New York, NY

Course Assistant, Department of Computer Science

Sept. 2024 - Dec 2025

- Held office hours and graded assignments for CS 4995: Randomized Algorithms and COMS 6998: Algorithms for Massive Data.

Flatiron Institute - Simons Foundation

New York, NY

Research Intern, Neural Circuits and Algorithms Group

May 2024 - Aug. 2024

- Redesigned CaImAn's large-scale calcium imaging data analysis tool from TensorFlow using PyTorch and Keras.
- Reintegrated an automated and scalable analysis pipeline for voltage imaging datasets with a PyTorch backend.
- Utilized Flatiron Institute's HPC cluster to train, test, and execute PyTorch-implemented supervised machine learning models on 10+ TB calcium imaging datasets. Code predominantly in Python. Coordinated with Janelia scientists.

Flatiron Institute - Simons Foundation

New York, NY

Research Intern, Neural Circuits and Algorithms Group

Sept. 2022 - Aug 2023

- Developed and evaluated a supervised IsoRSUNet architecture to improve neuron-boundary segmentation in assembled connectome maps. Conducted ablation experiments to analyze architectural stability and generalization.
- Leveraged Flatiron Institute's HPC cluster to automate segmentation processes. Code predominantly in Python.

Selected Research and Technical Projects

ICML HiLD Workshop 2026, "Rank-One Spectral Normalization Accelerates Projected Feature Learning"

- Analyzed and validated optimization dynamics through finite-width student-teacher neural-network experiments.

Undergraduate Project in CS, "Quantum Algorithm for Detecting Gravitational Waves"

Nov. 2022 - May 2023

- Executed in Python an innovative quantum advantage pattern recognition method on LIGO gravitational-wave data.

Leadership

Columbia Undergraduate Quantum Computing Club

Nov. 2022 - Jan. 2023

- Co-founder and former co-head; organized seminars, technical events, and managed weekly member communications

Awards

MIT IQuHACK Hackathon 2023; Covalent x IBM Challenge - 1st place

Jan 2023

Simon Foundation Global Brain SURF Fellowship

Aug 2022

Technical Skills

Languages and Frameworks: Python, C/C++, Java, MATLAB, R, PyTorch, TensorFlow, Jupyter, Keras, Slurm, Qiskit